

BakerySense

A self-evolving operations harness **for bakeries and perishable SMEs.**

It turns sales history into tomorrow's bake plan — then learns from its own decisions, evolving a different playbook for every shop, under one-tap human approval.

THE PROBLEM

30-40%

of everything an independent bakery bakes is thrown out.

The waste isn't ignorance — it's uncertainty.

Forecasts are never perfectly calibrated. Every branch behaves differently. No busy owner notices that croissants quietly run 25% high every Wednesday at one shop, and every Sunday at another. That blind spot is pure margin on the floor.

WHAT IT DOES

Tomorrow's bake plan, as a decision — not a chatbot

BakerySense

DashboardChatDisplay caseAdmin

Bukit Bintangno connectorSettingsSign out

TENANT - DEMO

Today's bake plan

Close out todayAsk Gemma for a narrative →

Branch brn_fxjy1wpy7y6 · 2026-05-29

THREE PLANS PER SKU

Pick a plan to commit. Numbers are expected outcomes from the quantile forecast under your cost ratio.

CROISSANT

CONSERVATIVE75

BALANCED70

AGGRESSIVE121

PAIN AU CHOCOLAT

CONSERVATIVE70

BALANCED60

AGGRESSIVE121

TRADITIONAL BAGUETTE

CONSERVATIVE108

BALANCED140

AGGRESSIVE121

Three plans per SKU.
Conservative · Balanced · Aggressive — each with expected waste, stockout chance, and units sold, picked by newsvendor math from a quantile forecast under the shop's own cost ratio. One tap commits the call.

It learns from its own decisions

1

diagnose

Classify every miss.
Stockout-censored,
operator override, one-off
event, or a genuine skill
fault — only the last one is
learnable.

2

propose

Emit a bounded, learning-
rate-capped correction to a
version-controlled skill
artifact (not the model).

3

validate

Re-score on a strictly held-
out window. The edit must
measurably cut error before
anyone sees it.

4

approve

The owner approves in one
tap. Controlled autonomy
— every edit is validated
and auditable.

The forecasting model stays frozen. The skills evolve.

SELF-INSPECTION IN ACTION

Corrections it taught itself, waiting for review

BakerySense

Dashboard

Chat

Display case

Admin

Bukit Bintang

no connector

Settings

Sign out

Connectors

Data

Users

Branches

Model

Harness

Audit

TENANT · DEMO

Harness evolution

Corrections the harness has learned from this tenant's own sales, each validated on a held-out window. Approving an item activates a new skill version; the next forecast applies it.

RUN INSPECTION

Replay this branch's recent traces, diagnose misses, and propose a validated correction.

Bukit Bintang

Run inspection

forecast · Subang Jaya

WAPE 3.4% → 2.2%

CROISSANT on Sun: forecast over by ~23% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Sun: 1 → 0.8

Approve

Reject

forecast · Bukit Bintang

WAPE 3.0% → 2.0%

CROISSANT on Wed: forecast over by ~20% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Wed: 1 → 0.8

Approve

Reject

Every night, the harness replays its traces and proposes validated fixes.

Bukit Bintang: croissants +25% on Wednesdays.
Held-out WAPE 3.0% → 2.0%.

CROISSANT · Wed 1.00 → 0.80

Same brand, same model — different playbooks

Bukit Bintang

forecast · Bukit Bintang

WAPE 3.0% → 2.0%

CROISSANT on Wed: forecast over by ~20% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Wed: 1 → 0.8

Approve

Reject

CROISSANT · Wed → 0.80

learned Wednesdays on its own

Subang Jaya

forecast · Subang Jaya

WAPE 3.4% → 2.2%

CROISSANT on Sun: forecast over by ~23% across 8 days → lower multiplier 1 → 0.8

CROISSANT on Sun: 1 → 0.8

Approve

Reject

CROISSANT · Sun → 0.80

learned Sundays on its own

Each shop becomes its own agent with its own skill set — branch skills are sparse overrides on brand defaults. A federation of self-evolving shops.

EXECUTION OVER POLISH

A working loop on live infrastructure — not a mockup

285

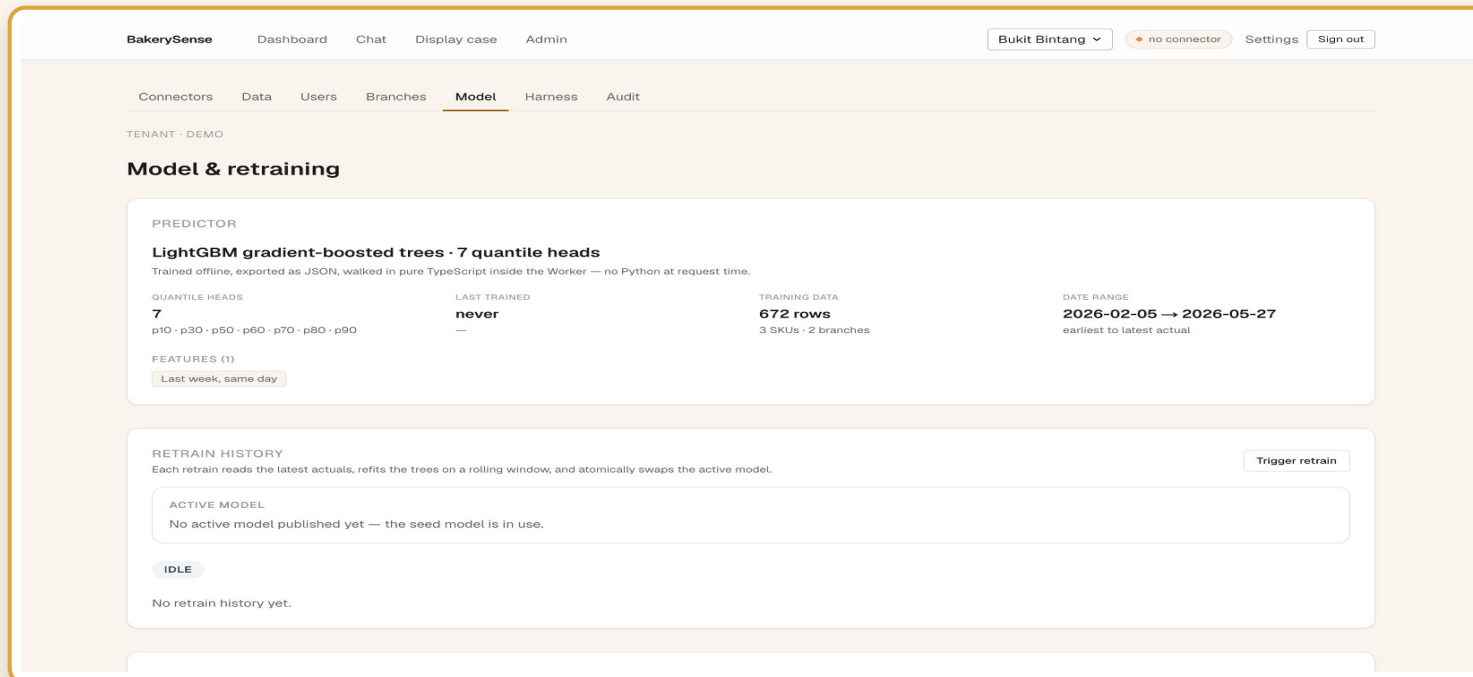
tests passing

100%

deterministic numbers

Live

on Cloudflare D1



Deterministic numbers, semantic Gemma.

Quantities come from a LightGBM quantile forecaster + newsvendor — never hallucinated. Gemma 4 is the narration and explanation layer. That split is what makes recommendations trustworthy to a real operator — and it's what gets the harness's edits past validation instead of vibes.

GLOBAL SCALABILITY

One harness, every perishable business

● Bakeries

● Cafés

● Grocers

● Florists

● Cloud kitchens

● Central kitchens

The brand→branch skill hierarchy scales to chains. Only **learned rule diffs** propagate — not raw sales — so it preserves each shop's data privacy while sharing what works.

Real product value · execution quality · global scalability.

GET STARTED

Try it in 30 seconds

Live demo `bakerysense.swmengappdev.workers.dev`

Sign in `demo@bs.co` / `Password2026Password` (slug: demo)

Money shot Admin → Harness — two branches, two different learned fixes, awaiting approval.

Repo

`github.com/Whyme-Labs/bakerysense`